Information Collection Strategies Experiment

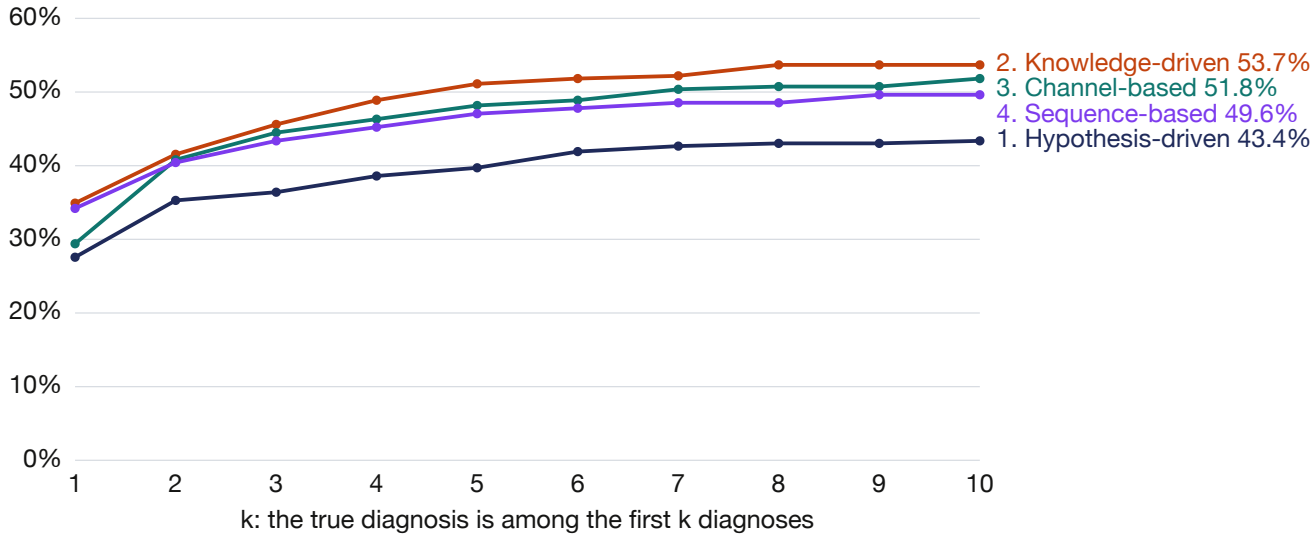
Diagnostic accuracy of four information collection strategies on 272 simulated cases

Run 2026-10-19T19:40:58+00:00 to 2026-10-19T22:49:10+00:00 · record `strategies_272_16it.json` · status completed · commit 7e10b23

Result

Strategy	Hit rate@1	Hit rate@3	Hit rate@5	Hit rate@10	nDCG@10	MRR	Cases
1. Hypothesis-driven	27.6%	36.4%	39.7%	43.4%	0.356	0.331	272
2. Knowledge-driven	**34.9%**	**45.6%**	**51.1%**	**53.7%**	**0.443**	**0.412**	272
3. Channel-based	29.4%	44.5%	48.2%	51.8%	0.411	0.377	272
4. Sequence-based	34.2%	43.4%	47.1%	49.6%	0.419	0.395	272

Bold marks the best value in a column. Hit rate@k: share of cases whose true diagnosis code is among the first k codes of the strategy's differential. nDCG@10 and MRR reward a true diagnosis placed higher. Every case counts in the denominator.



How sure the differences are

Strategy	Hit rate@1 (95% interval)	Hit rate@10 (95% interval)
1. Hypothesis-driven	27.6% (22.6% to 33.2%)	43.4% (37.6% to 49.3%)
2. Knowledge-driven	34.9% (29.5% to 40.8%)	53.7% (47.7% to 59.5%)
3. Channel-based	29.4% (24.3% to 35.1%)	51.8% (45.9% to 57.7%)
4. Sequence-based	34.2% (28.8% to 40.0%)	49.6% (43.7% to 55.5%)

Wilson intervals over the cases of this single run. The strategies ran on the same cases, so the paired comparison below is the sharper test.

Pair (first vs second)	@1: cases only the first / only the second hit	p	@10: only the first / only the second	p
1. Hypothesis-driven vs 2. Knowledge-driven	17 / 37	0.009	20 / 48	**0.001**
1. Hypothesis-driven vs 3. Channel-based	34 / 39	0.640	30 / 53	0.015
1. Hypothesis-driven vs 4. Sequence-based	27 / 45	0.044	29 / 46	0.064
2. Knowledge-driven vs 3. Channel-based	40 / 25	0.082	41 / 36	0.649
2. Knowledge-driven vs 4. Sequence-based	40 / 38	0.910	39 / 28	0.222
3. Channel-based vs 4. Sequence-based	29 / 42	0.154	41 / 35	0.567

Exact McNemar test on the same 272 cases. With 6 comparisons, a p-value below 0.008 (bold) holds after a Bonferroni correction; values between that and 0.05 are suggestive only.

What was run

- 272 cases from `cases/combined_272_whole_chunking.json`, each run with every strategy: 1088 case runs.
- Each strategy could ask at most 16 questions and could stop earlier by its own rule. When it stopped, it gave a differential of up to 10 diagnoses with probabilities.
- Strategy model: anthropic/claude-haiku-5.5. Simulated environment model: openai/gpt-6-luna; literature sources europe_pmc, litsense.
- Coding: ICD-11 MMS 2024-01, first result of the flexible search for the diagnosis name. A prediction counts as a hit only when its code equals the true diagnosis code exactly.
- Total time 3.1 h; total cost $25.70 (strategies $12.48, environment $13.22).

How the strategies behaved

Strategy	Questions per case	Why runs stopped	Strategy LLM calls per case	Seconds per case	Strategy cost	Environment cost	Failed runs	No differential
1. Hypothesis-driven	13.18	max iterations 170, strategy done 102	27.0	624	$4.56	$3.08	0	0
2. Knowledge-driven	15.94	max iterations 267, strategy done 5	32.0	686	$4.39	$3.61	0	0
3. Channel-based	14.79	max iterations 229, strategy done 43	32.1	680	$1.39	$3.43	0	0
4. Sequence-based	13.26	strategy done 199, max iterations 73	24.3	599	$2.15	$3.11	0	0

Where the answers came from (questions)	Case study	Literature	Unanswerable	No answer (error)
1. Hypothesis-driven	676	2750	159	0
2. Knowledge-driven	927	3228	181	0
3. Channel-based	676	3092	254	0
4. Sequence-based	686	2741	180	0

Coding

Strategy	Predicted diagnoses	Without an ICD code	Cases with no coded prediction	Cases whose true diagnosis has no code
1. Hypothesis-driven	2720	4	0	2
2. Knowledge-driven	2720	3	0	2
3. Channel-based	2720	1	0	2
4. Sequence-based	2720	2	0	2

4684 distinct diagnosis names were coded; 7 have no ICD-11 match. A case whose true diagnosis has no code cannot be hit by any strategy.

Reading these numbers

- Exact code match is strict: a correct diagnosis worded so that the search returns a neighbouring code (a parent, a sibling, or a code with a different extension) counts as a miss.
- One run per case and strategy. LLM sampling differs between runs, so small differences between strategies may not repeat.
- The stopping thresholds are starting values and were not tuned.
- Answers marked "literature" are synthesized by the simulated environment when the case text is silent; they are plausible, not facts about the patient.